

A REVIEW OF IMBALANCED DATA PROCESSING METHODS: A CASE STUDY ON TOXICITY DETECTION IN SOCIAL MEDIA

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Team Member

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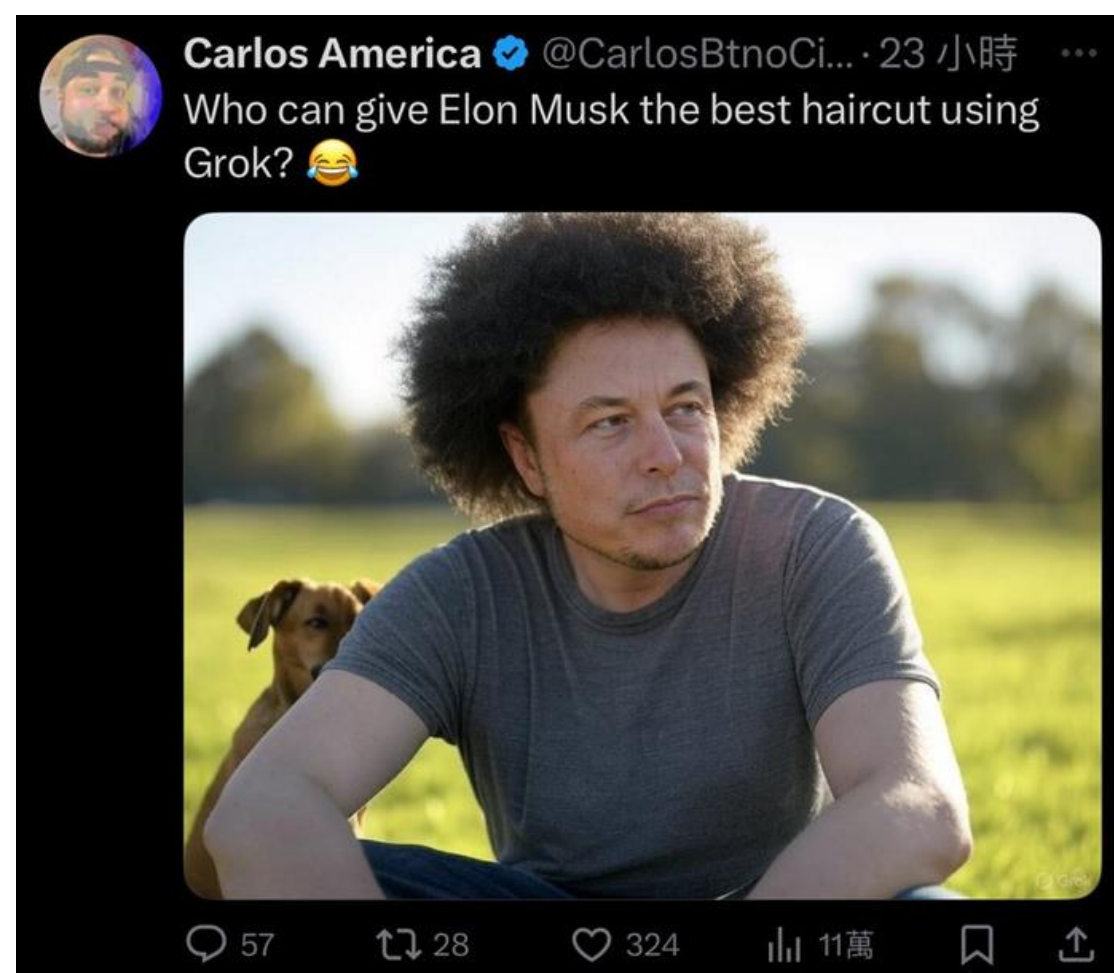
113423068 Rogers Zheng



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BACKGROUND



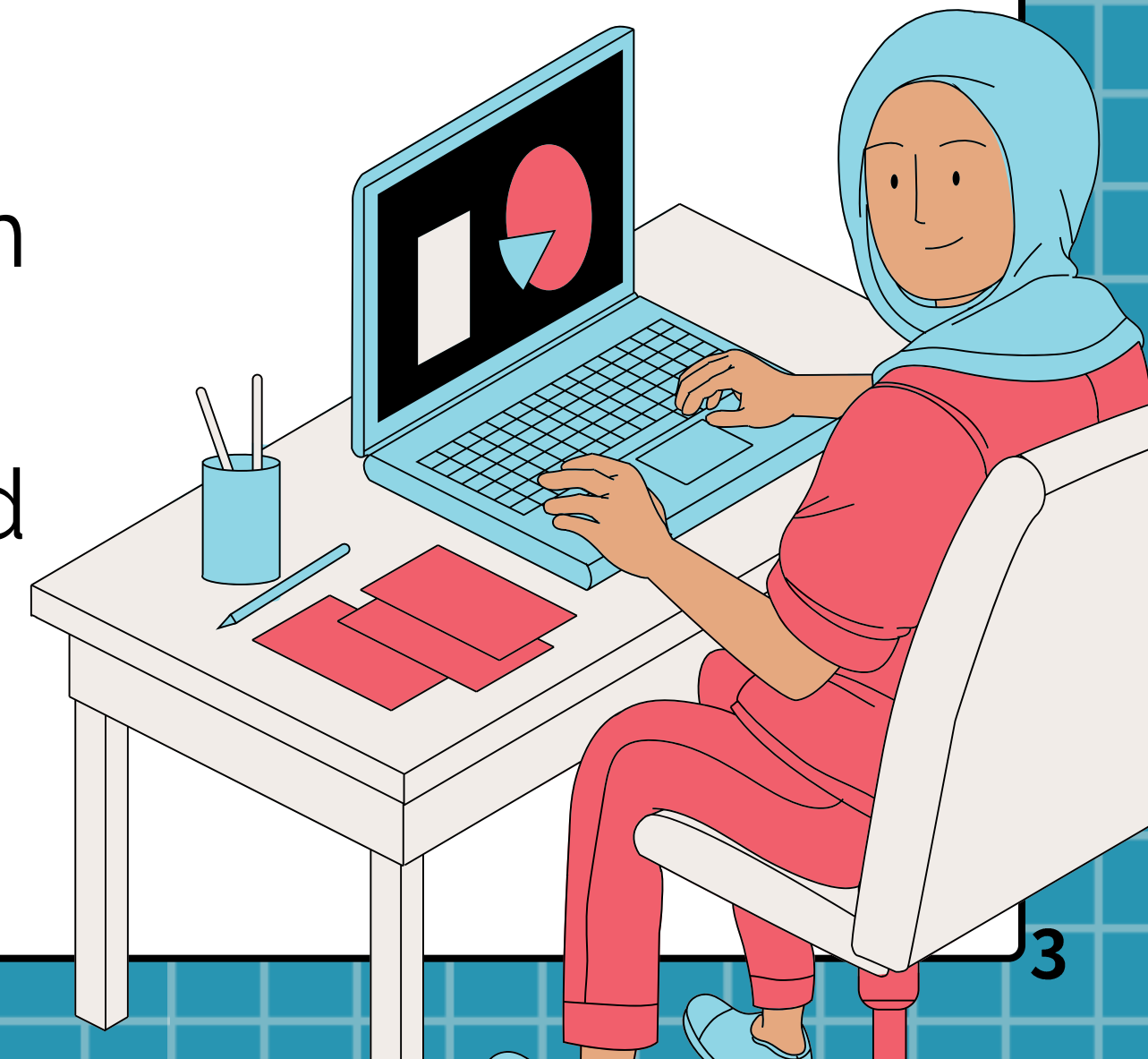
BACKGROUND

Recently, toxic discourse has been frequently occurring on social media platforms such as YouTube, X, and Threads, with a surge in dangerous and harmful comments. To ensure community safety and protect minors, it is crucial to promptly remove these extreme and harmful remarks. However, the proportion of harmful content is significantly different from that of normal content, making detection challenging. Therefore, we have designed a series of experiments to verify and enhance the identification of such harmful discourse.

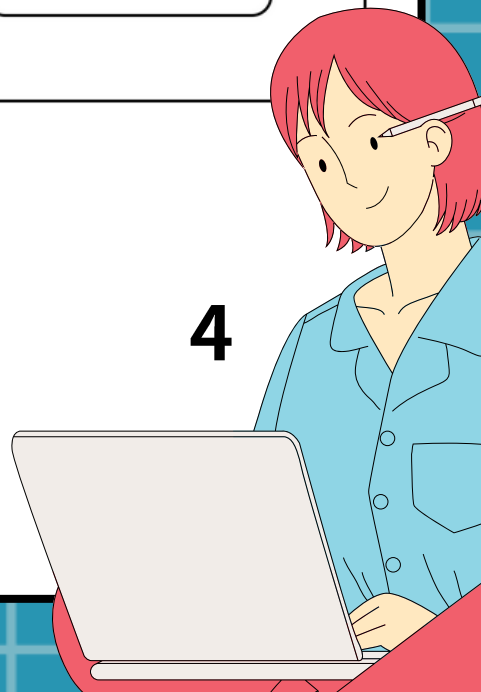
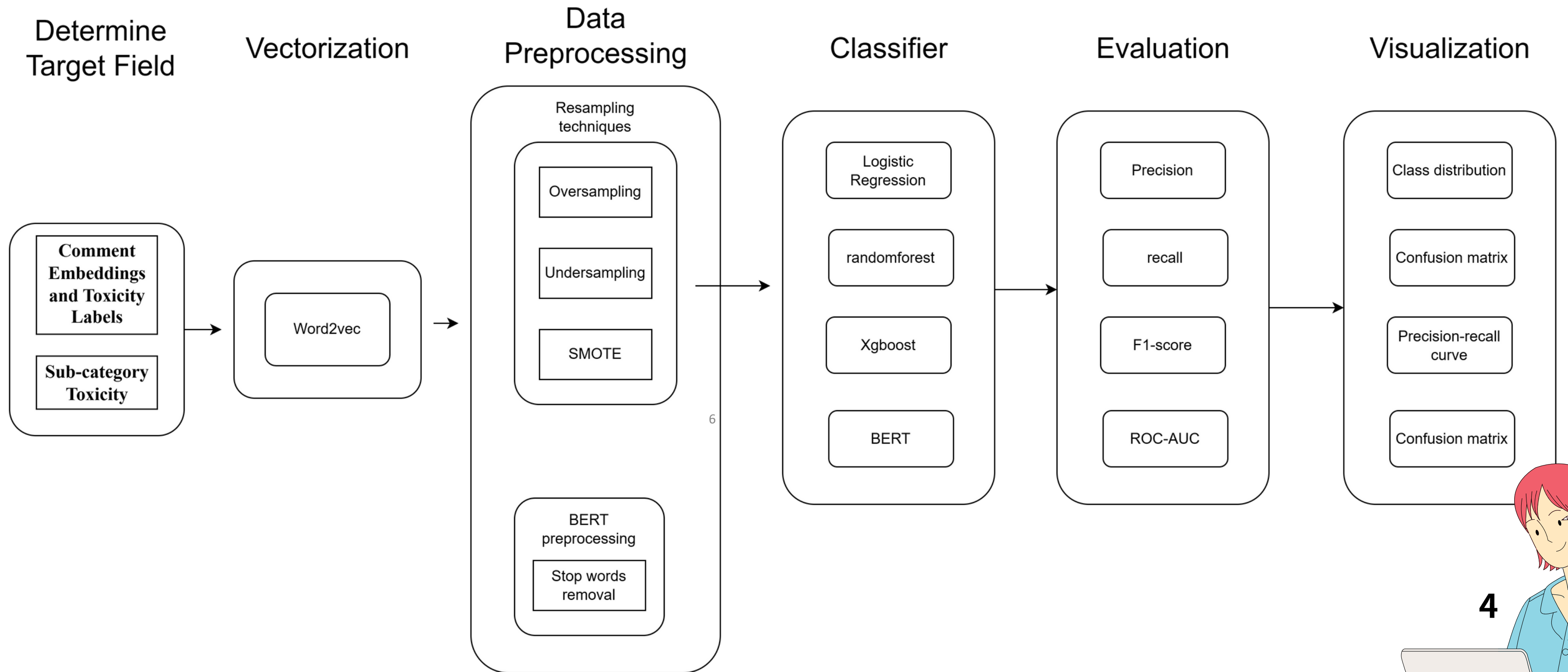
OBJECTIVES AND CONTRIBUTIONS

- Explore different methods for dealing with imbalanced data (By data level and model level)
- Take social media toxicity testing as an Example
- Provide experimental comparison and indicator evaluation

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EXPERIMENT FLOWCHART



DATASET

	A	B	C	D	E	F	G	H	I	J
1	comment_text	id	identity_hate	insult	obscene	set	severe_toxic	threat	toxic	toxicity
2	they weren t vandalism just closure on some gas after i voted at new yo	0000997932d777bf	0	0	0	train	0	0	0	0
3	matches this background colour i m seemingly stuck with thanks talk j	000103f0d9cfb60f	0	0	0	train	0	0	0	0
4	removing relevant information and talking to me through edits instead of	000113f07ec002fd	0	0	0	train	0	0	0	0
5	that later on if no one else does first if you have any preferences for form	0001b41b1c6bb37e	0	0	0	train	0	0	0	0
6	you sir are my hero any chance you remember what page that s on	0001d958c54c6e35	0	0	0	train	0	0	0	0
7	congratulations from me as well use the tools well talk	00025465d4725e87	0	0	0	train	0	0	0	0
8	cock suck before you piss around on my work	0002bcb3da6cb337	0	1	1	train	1	0	1	4
9	he matt shirvington article has been reverted please don t do it again or	00031b1e95af7921	0	0	0	train	0	0	0	0
10	encyclopedia so one can use it for school as a reference i have been to t	00037261f536c51d	0	0	0	train	0	0	0	0
11	alignment on this subject and which are contrary to those of dulithgow	00040093b2687caa	0	0	0	train	0	0	0	0
12	the image description page and edit it to include a fair use rationale if yo	0005300084f90edc	0	0	0	train	0	0	0	0
13	bbq be a man and lets discuss it maybe over the phone	00054a5e18b50dd4	0	0	0	train	0	0	0	0
14	tourist who gang up any one who asks them questions abt their anti social	0005c987bdfc9d4b	0	0	0	train	0	0	1	1
15	he is the only living ex president who did not attend that s certainly more	0006f16e4e9f292e	0	0	0	train	0	0	0	0
16	where it doesn t belong i believe the argument was between me and yve	00070ef96486d6f9	0	0	0	train	0	0	0	0
17	a singles santana s town and down so yes he is born in he really is how	00078f8ce7eb276d	0	0	0	train	0	0	0	0
18	bye don t look come or think of coming back tosser	0007e25b2121310b	0	0	0	train	0	0	1	1
19	redirect talk voydan pop georgiev chernodrinski	000897889268bc93	0	0	0	train	0	0	0	0
20	de no sense why not argue to include hindi on ryo sakazaki s page to inc	0009801bd85e5806	0	0	0	train	0	0	0	0
21	then even better i d like to ask you to take a closer look at the premature v	0009eaea3325de8c	0	0	0	train	0	0	0	0
22	pre film articles your edits are simply not good with entirely too many u	000b08c464718505	0	0	0	train	0	0	0	0
23	good to know about me yeah i m studying now deepu	000bfd0867774845	0	0	0	train	0	0	0	0
24	are by far the most common variety http w its caltech edu atomic snow	000c0dfd995809fa	0	0	0	train	0	0	0	0
25	he signpost september read this signpost in full single page unsubscribe	000c6a3f0cd3ba8e	0	0	0	train	0	0	0	0
26	reverted i would invite anyone to re visit articles written about the followi	000cfee90f50d471	0	0	0	train	0	0	0	5
27	cluded in the echinodermata were bilateral such as homostelea or even a	000eefc67a2c930f	0	0	0	train	0	0	0	0

DATASET

Use comment to predict the following six columns

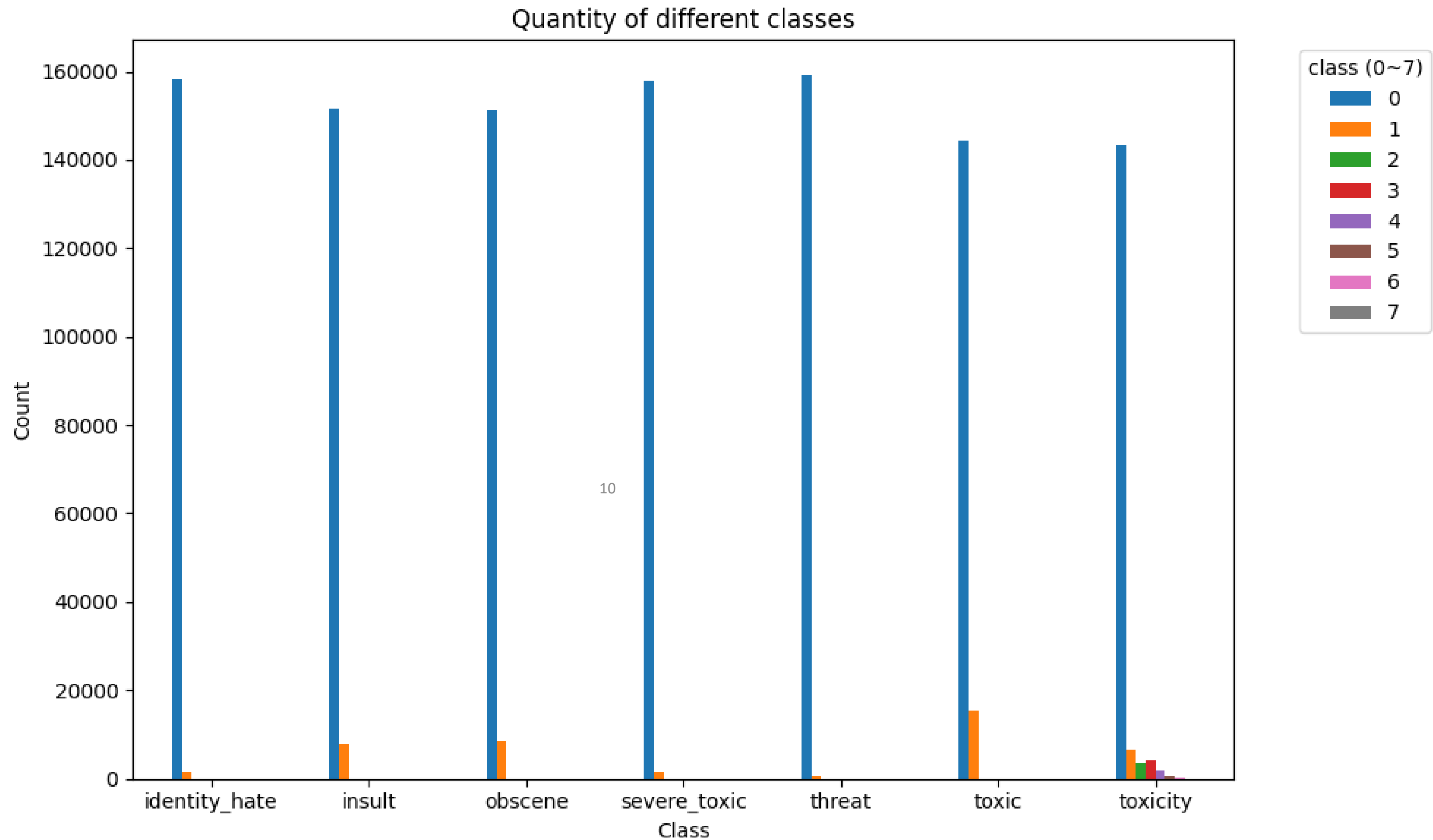
	A	B	C	D	E	F	G
1	comment_text	identity_hate	insult	obscene	severe_toxic	threat	toxic
2	explanation why the edits made under my username hardcore metallica fan were r	0	0	0	0	0	0
3	d aww he matches this background colour i m seemingly stuck with thanks talk	0	0	0	0	0	0
4	hey man i m really not trying to edit war it s just that this guy is constantly remov	0	0	0	0	0	0
5	more i can t make any real suggestions on improvement i wondered if the section	0	0	0	0	0	0
6	you sir are my hero any chance you remember what page that s on	0	0	0	0	0	0
7	congratulations from me as well use the tools well talk	0	0	0	0	0	0
8	cock suck before you piss around on my work	0	1	1	1	0	1
9	your vandalism to the matt shirvington article has been reverted please don t do it	0	0	0	0	0	0
10	sorry if the word nonsense was offensive to you anyway i m not intending to w	0	0	0	0	0	0
11	alignment on this subject and which are contrary to those of dulithgow ⁸	0	0	0	0	0	0
12	fair use rationale for image wonju jpg thanks for uploading image wonju jpg i r	0	0	0	0	0	0
13	bbq be a man and lets discuss it maybe over the phone	0	0	0	0	0	0
14	hey what is it talk what is it an exclusive group of some wp talibans who are goo	0	0	0	0	0	1
15	before you start throwing accusations and warnings at me lets review the edit itse	0	0	0	0	0	0
16	oh and the girl above started her arguments with me she stuck her nose where it	0	0	0	0	0	0
17	juelz santanas age in juelz santana was years old then came february th which r	0	0	0	0	0	0
18	bye don t look come or think of comming back tosser	0	0	0	0	0	1
19	redirect talk voydan pop georgiev chernodrinski	0	0	0	0	0	0
20	the mitsurugi point made no sense why not argue to include hindi on ryo sakazaki	0	0	0	0	0	0

DATASET

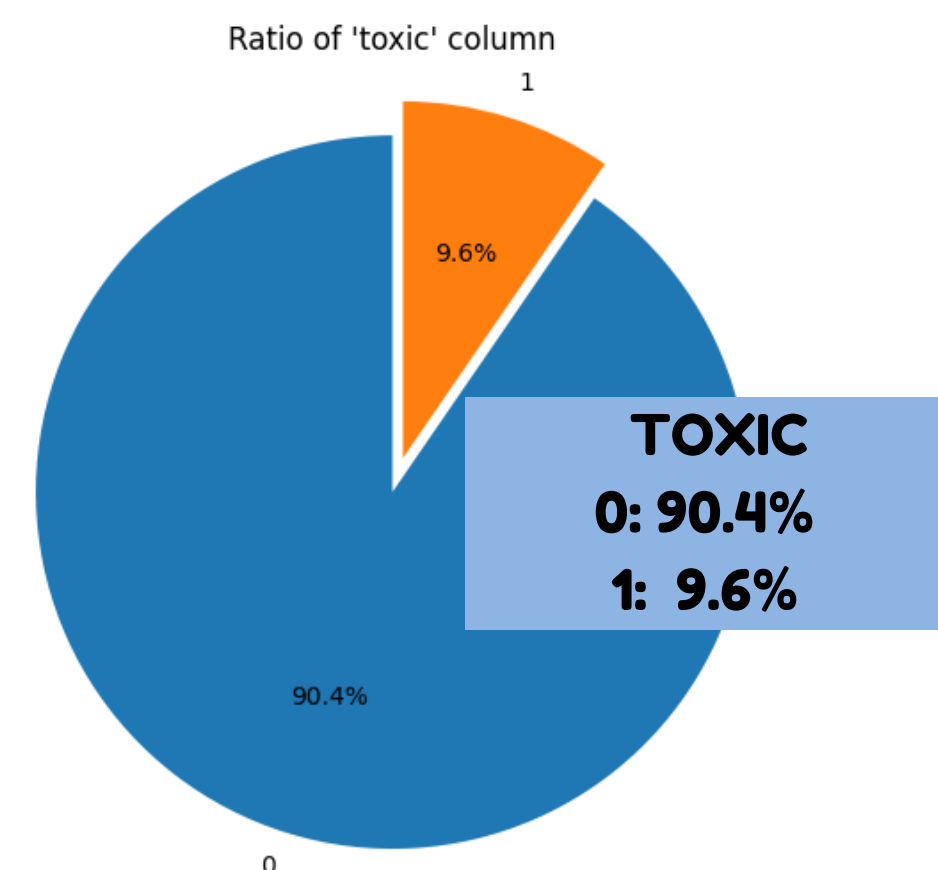
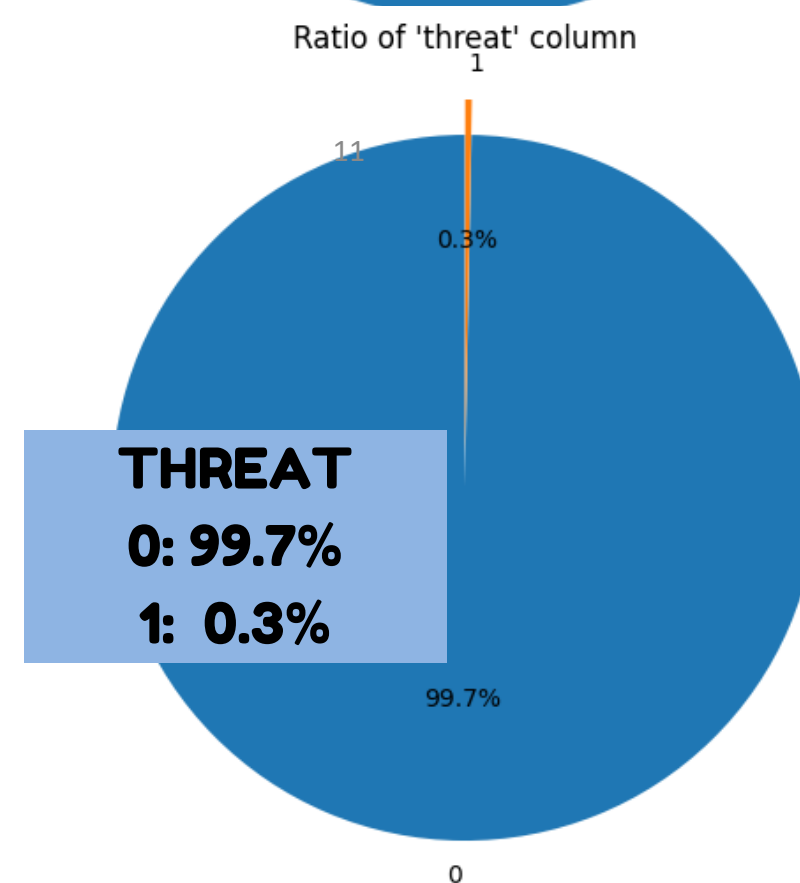
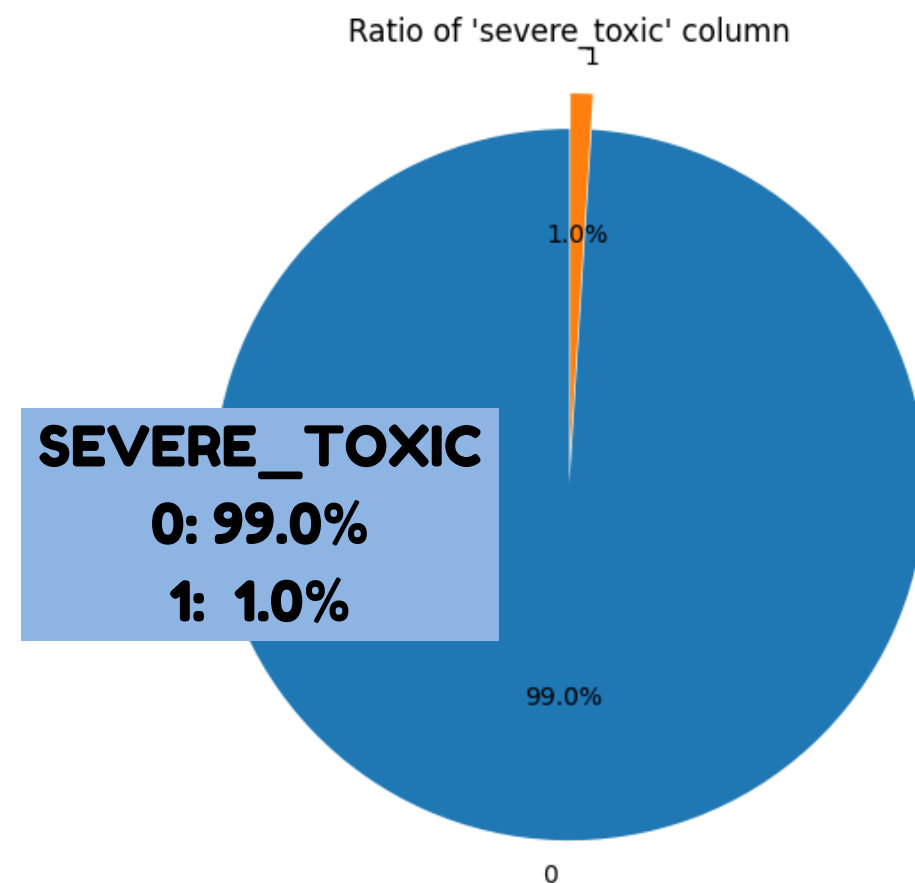
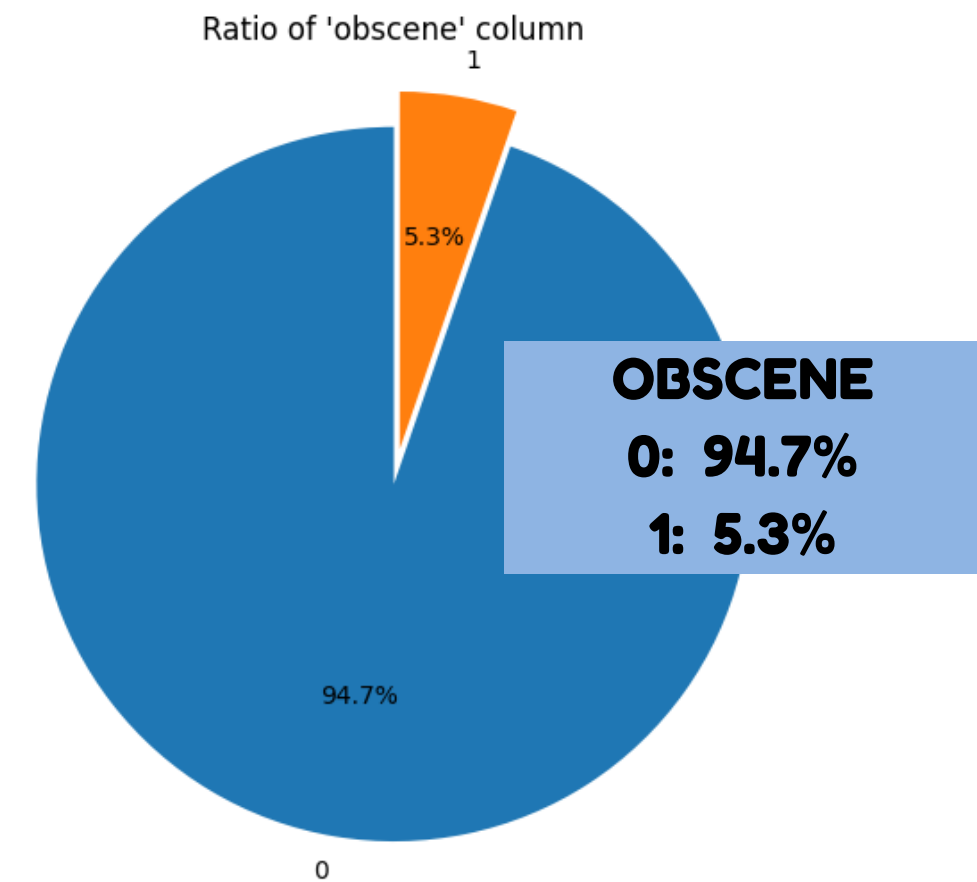
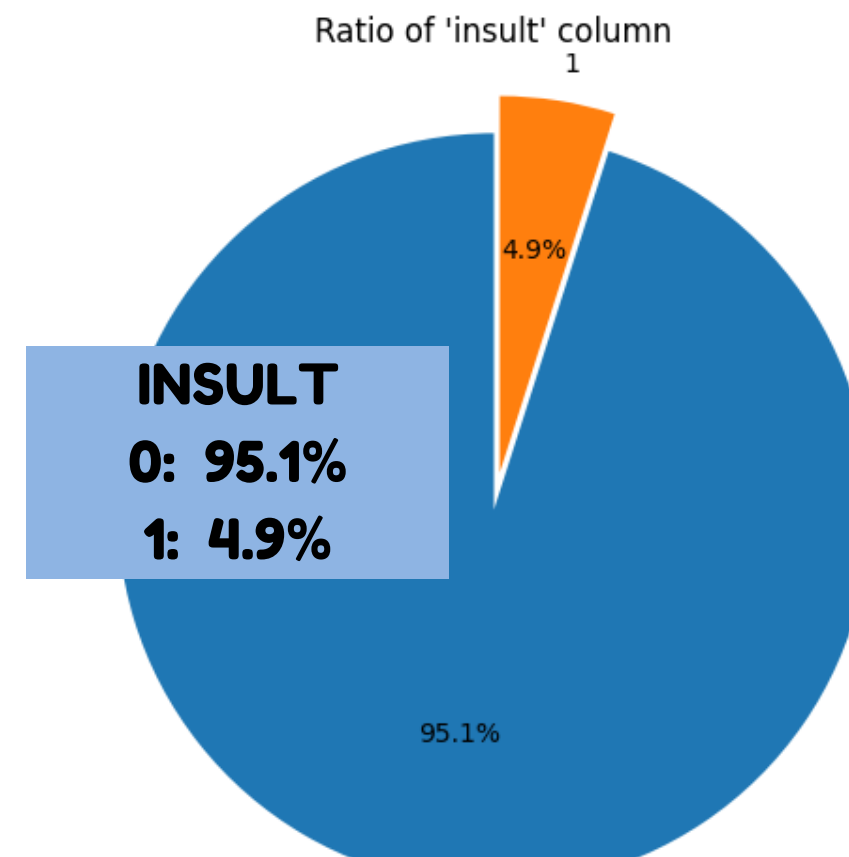
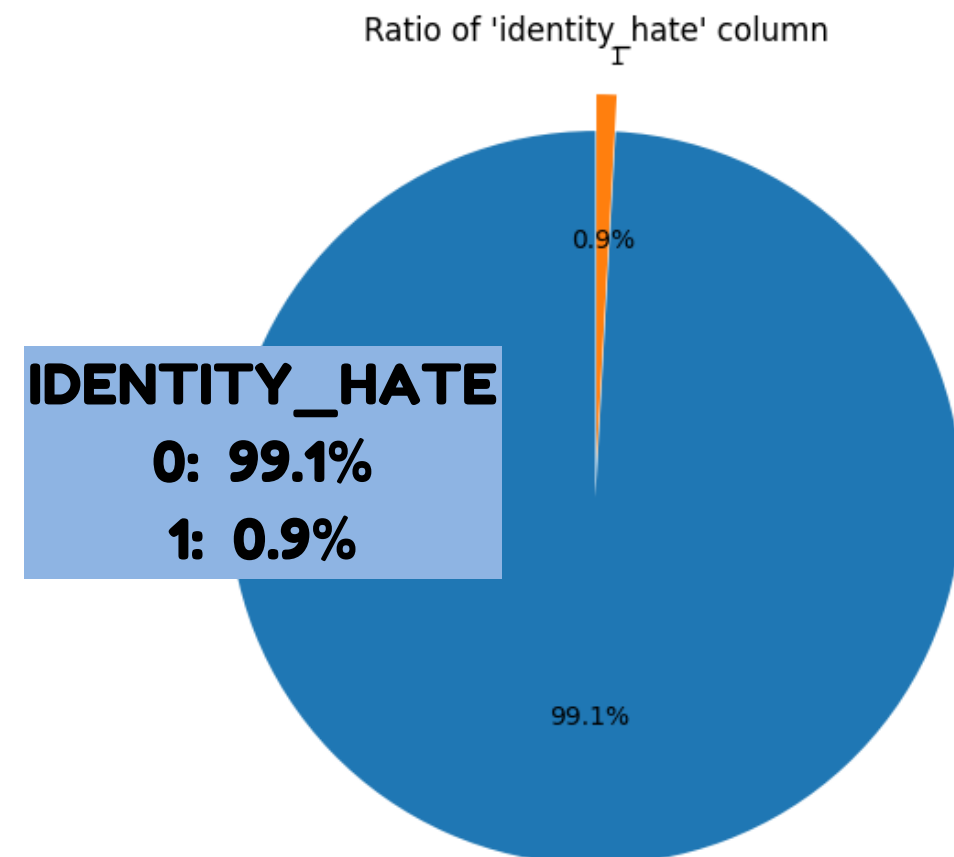
Use comment to predict the toxicity

	A	B
1	comment_text	toxicity
2	explanation why the edits made under my username hardcore metallica fan were	0
3	d aww he matches this background colour i m seemingly stuck with thanks talk	0
4	hey man i m really not trying to edit war it s just that this guy is constantly remo	0
5	more i can t make any real suggestions on improvement i wondered if the sectio	0
6	you sir are my hero any chance you remember what page that s on	0
7	congratulations from me as well use the tools well talk	0
8	cock suck before you piss around on my work	4
9	your vandalism to the matt shirvington article has been reverted please don t do	0
10	sorry if the word nonsense was offensive to you anyway i m not intending to	0
11	alignment on this subject and which are contrary to those of dulithgow	0
12	fair use rationale for image wonju jpg thanks for uploading image wonju jpg i	0
13	bbq be a man and lets discuss it maybe over the phone	0
14	hey what is it talk what is it an exclusive group of some wp talibans who are gc	1
15	before you start throwing accusations and warnings at me lets review the edit its	0
16	oh and the girl above started her arguments with me she stuck her nose where i	0
17	juelz santanas age in juelz santana was years old then came february th which	0
18	bye don t look come or think of comming back tosser	1
19	redirect talk voydan pop georgiev chernodrinski	0

DATASET ANALYSIS

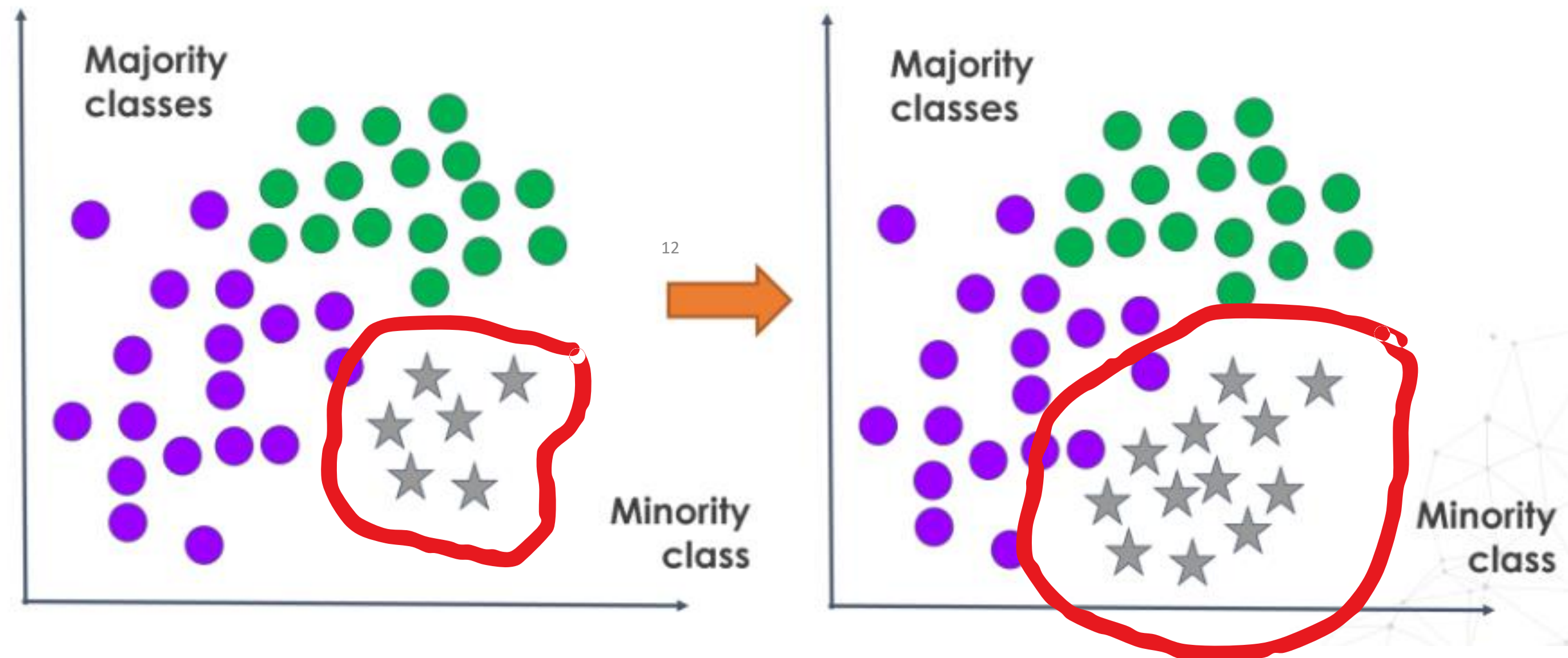


DATASET ANALYSIS



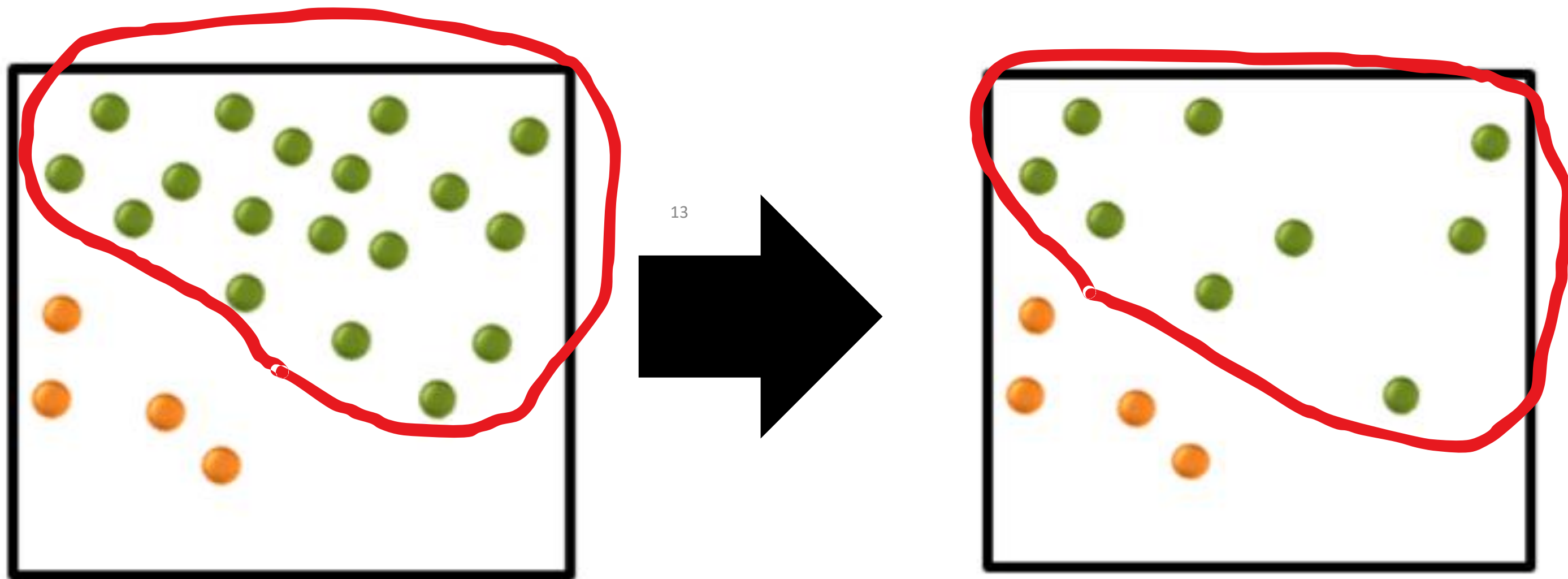
SAMPLING METHOD-OVERSAMPLING

Oversampling is the process of balancing class distribution in an imbalanced dataset by duplicating minority class samples or generating new synthetic samples to improve the model's ability to learn from the minority class.



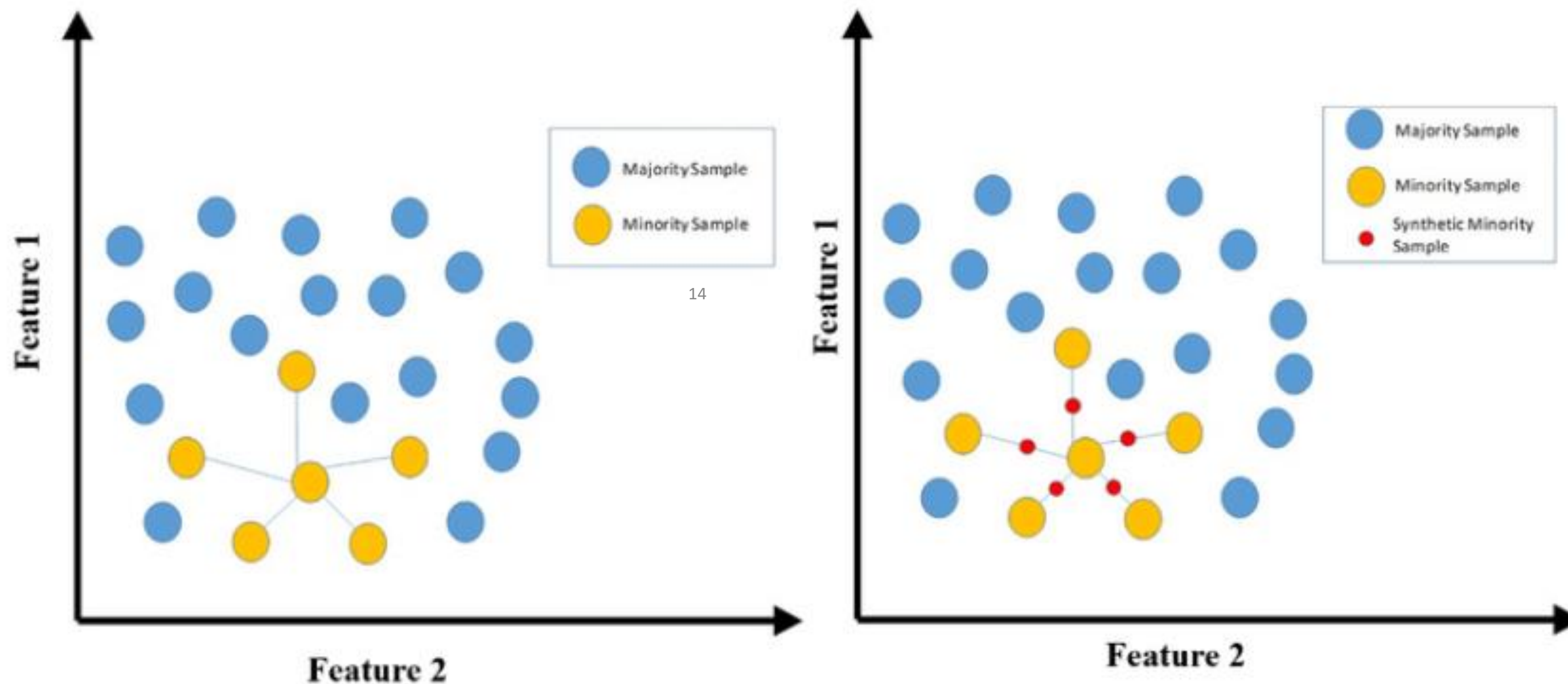
SAMPLING METHOD-UNDERSAMPLING

Undersampling is the process of balancing class distribution in an imbalanced dataset by reducing the number of majority class samples to mitigate model bias toward the majority class.



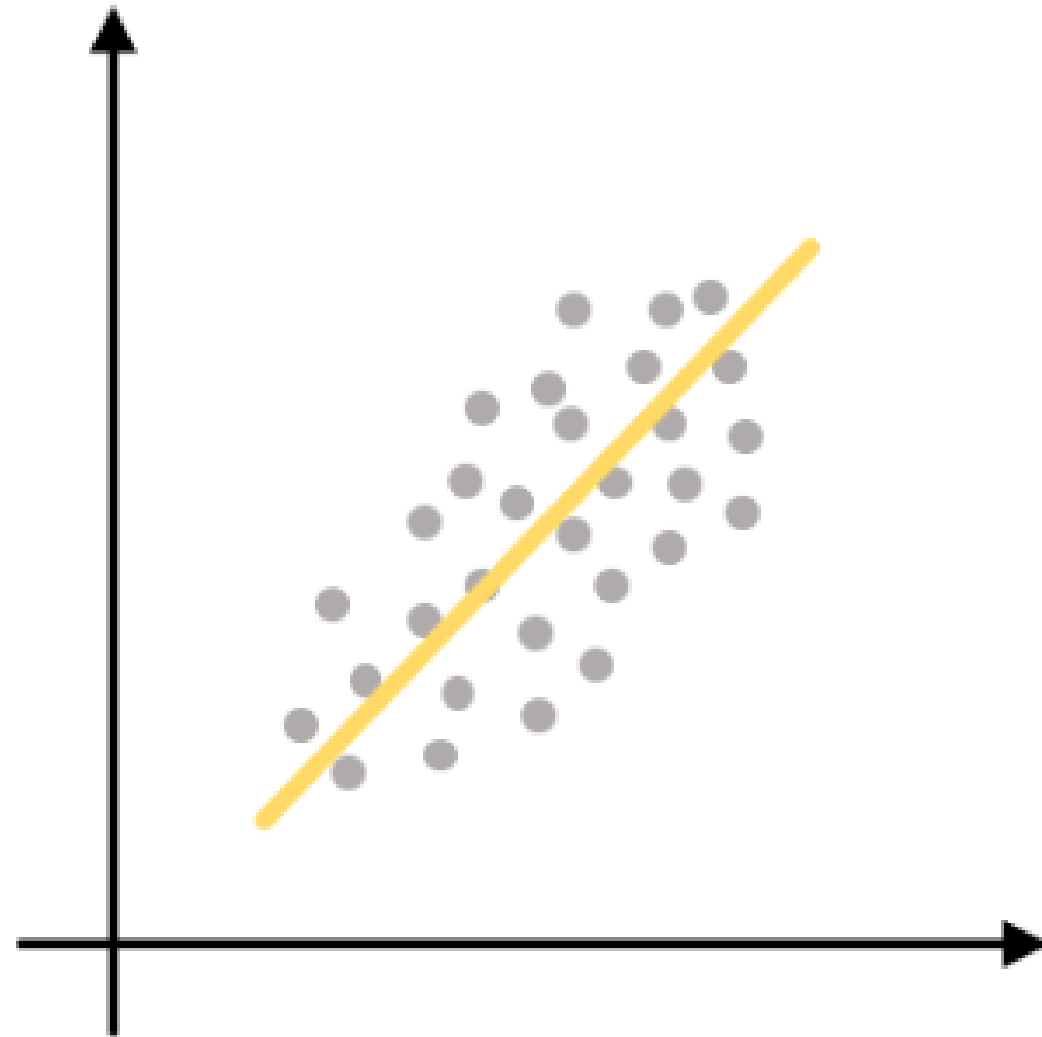
SAMPLING METHOD-SMOTE

SMOTE balances an imbalanced dataset by generating new synthetic minority class samples to improve the model's ability to learn from the minority class.



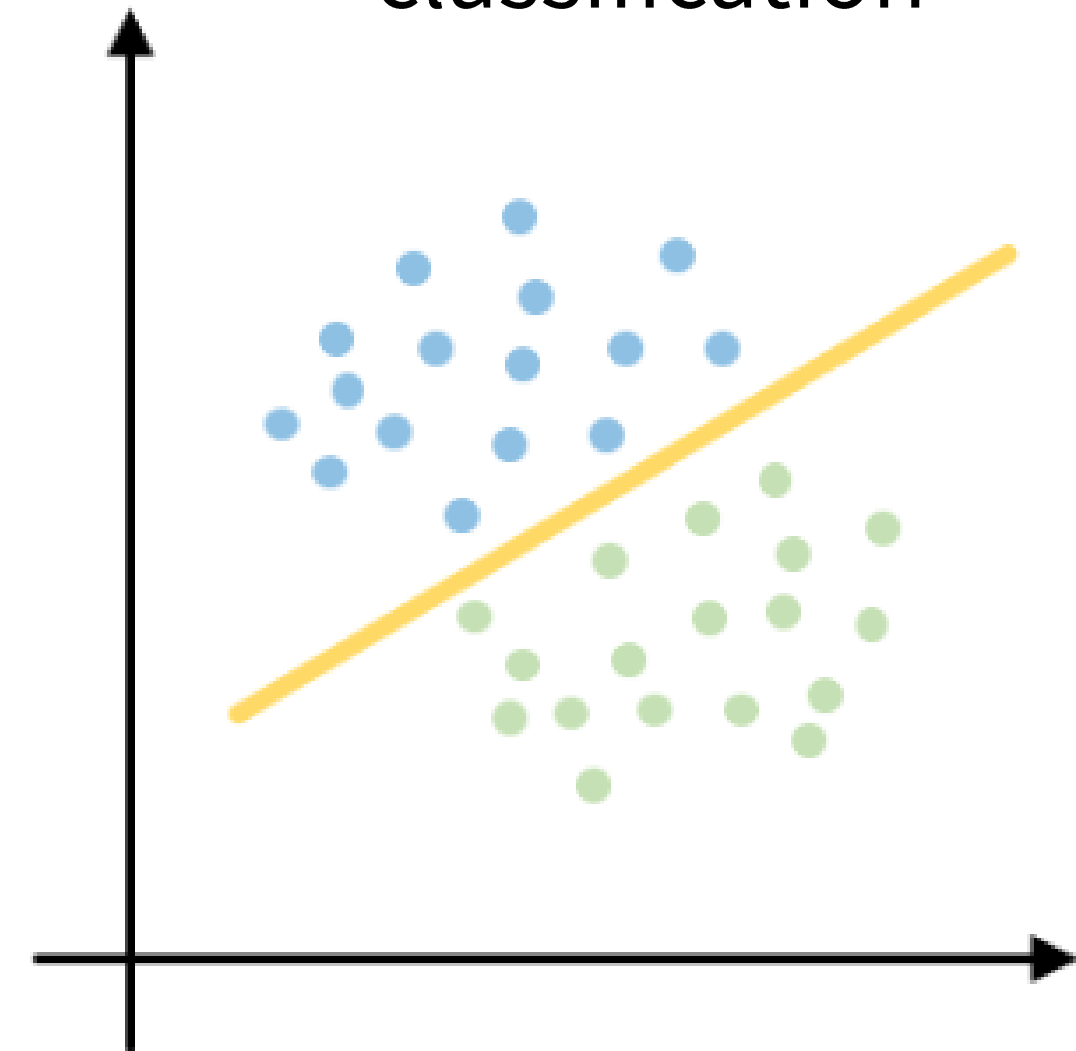
CLASSIFIER-LOGISTIC REGRESSION

Predicting
continuous values



LINEAR REGRESSION

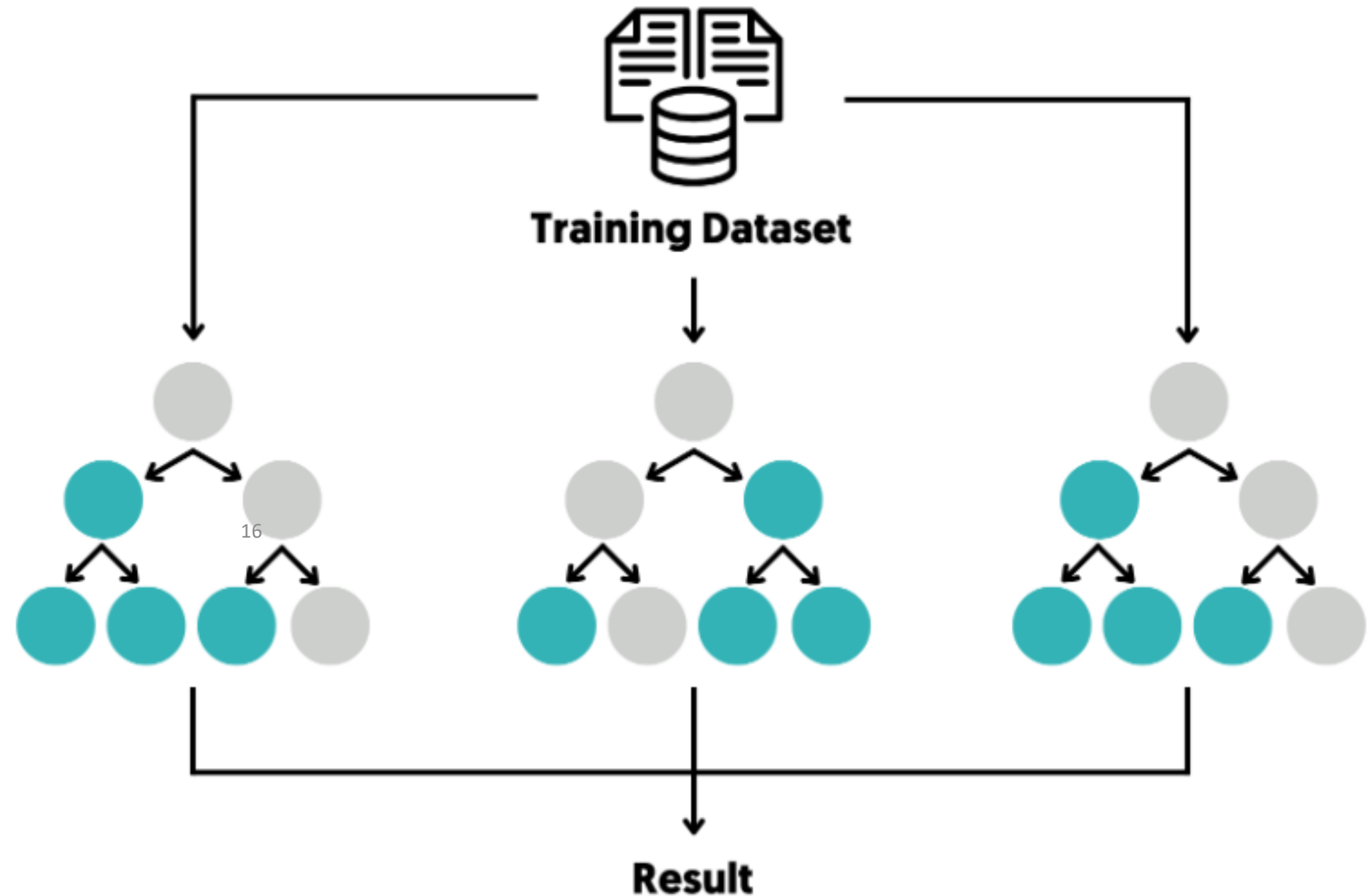
Binary or multi-class
classification



LOGISTIC REGRESSION

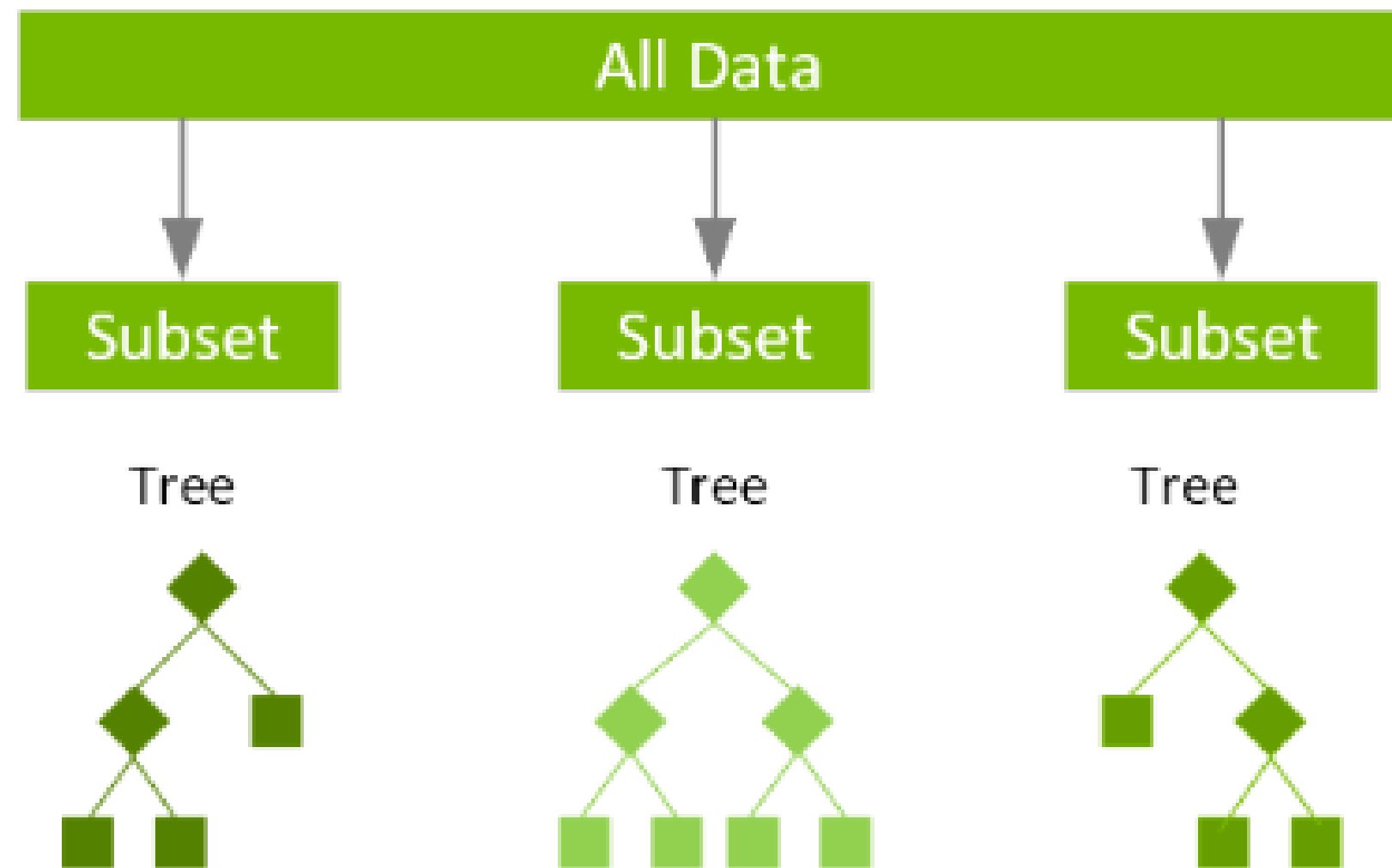
CLASSIFIER-RANDOM FOREST

Trains multiple decision trees and aggregates their outputs through voting or averaging to improve model accuracy and generalization.



CLASSIFIER-XGBOOST

A machine learning algorithm based on gradient boosting that enhances computational efficiency and generalization through parallel computing and regularization techniques.



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CLASSIFIER-BERT

BERT is a Transformer-based pre-trained language model that learns language representations through bidirectional context, achieving outstanding performance in natural language processing tasks.



EXPERIMENT RESULTS

IDENTITY_HATE

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.9989	0.9912	0.9887	0.9908	0.9909	0.9907	0.9894	0.9911	0.9897	0.9905	0.9911	0.9907
Recall	0.9915	0.9131	0.9867	0.9197	0.9177 ₂₀	0.9284	0.9996	0.9149	0.9842	0.9996	0.9149	0.9842
F1-score	0.9892	0.9472	0.9877	0.9508	0.9497	0.9556	0.9895	0.9482	0.9866	0.9717	0.9515	0.9757
AUC ROC	0.9371	0.9602	0.9652	0.9682	0.9677	0.9675	0.9628	0.9638	0.9617	0.9609	0.9704	0.9614

INSULT

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.9643	0.9611	0.9640	0.9636	0.9635	0.9630	0.9635	0.9613	0.9639	0.9641	0.9627	0.9650
Recall	0.9677	0.9167	0.9615	0.9256	0.9258 ₂₁	0.9292	0.9569	0.9121	0.9539	0.9271	0.9174	0.9392
F1-score	0.9648	0.9324	0.9626	0.9388	0.9389	0.9411	0.9596	0.9294	0.9578	0.9399	0.9332	0.9482
AUC ROC	0.9497	0.9555	0.9552	0.9642	0.9637	0.9615	0.9589	0.9615	0.9564	0.9698	0.9700	0.9686

OBSCENE

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.9735	0.9619	0.9725	0.9682	0.9677	0.9688	0.9752	0.9647	0.9738	0.9658	0.9642	0.9681
Recall	0.9751	0.9302	0.9727	0.9492	0.9479 ₂₂	0.9525	0.9738	0.9345	0.9707	0.9385	0.9299	0.9513
F1-score	0.9732	0.9412	0.9726	0.9555	0.9546	0.9580	0.9744	0.9446	0.9719	0.9477	0.9415	0.9482
AUC ROC	0.9659	0.9642	0.9707	0.9749	0.9758	0.9723	0.9761	0.9757	0.9743	0.9727	0.9718	0.9711

SEVERE_TOXIC

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.9873	0.9912	0.9893	0.9914	0.9914	0.9914	0.9890	0.9912	0.9910	0.9912	0.9911	0.9911
Recall	0.9889	0.9492	0.9842	0.9591	0.9557 ₂₃	0.9640	0.9862	0.9478	0.983	0.9693	0.9448	0.9724
F1-score	0.9880	0.9667	0.9864	0.9722	0.9704	0.9751	0.9748	0.9659	0.9862	0.9781	0.9642	0.9482
AUC ROC	0.9735	0.9828	0.9834	0.9856	0.9861	0.9846	0.9809	0.9850	0.9842	0.9794	0.9855	0.9797

THREAT

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.9958	0.9966	0.9955	0.9966	0.9967	0.9966	0.9962	0.9966	0.9960	0.9970	0.9970	0.9969
Recall	0.9970	0.9116	0.9952	0.9255	0.9067 ₂₄	0.9378	0.9970	0.9050	0.9943	0.9888	0.9446	0.9899
F1-score	0.9960	0.9510	0.9953	0.9585	0.9483	0.9652	0.9965	0.9473	0.9951	0.9923	0.9688	0.9482
AUC ROC	0.8781	0.9553	0.9522	0.9605	0.9620	0.9564	0.9316	0.9570	0.9327	0.9834	0.9853	0.9832

TOXIC

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.9450	0.9323	0.9425	0.9385	0.9381	0.9388	0.9434	0.9343	0.9399	0.9447	0.9426	0.9465
Recall	0.9485	0.8893	0.9414	0.9052	0.9033 ₂₅	0.9098	0.9337	0.8856	0.9271	0.9121	0.904	0.9235
F1-score	0.9441	0.9033	0.9419	0.9159	0.9144	0.9193	0.9373	0.9010	0.9318	0.9221	0.9158	0.9482
AUC ROC	0.9506	0.9473	0.9511	0.957	0.9565	0.9537	0.9550	0.9561	0.9538	0.9675	0.9665	0.9670

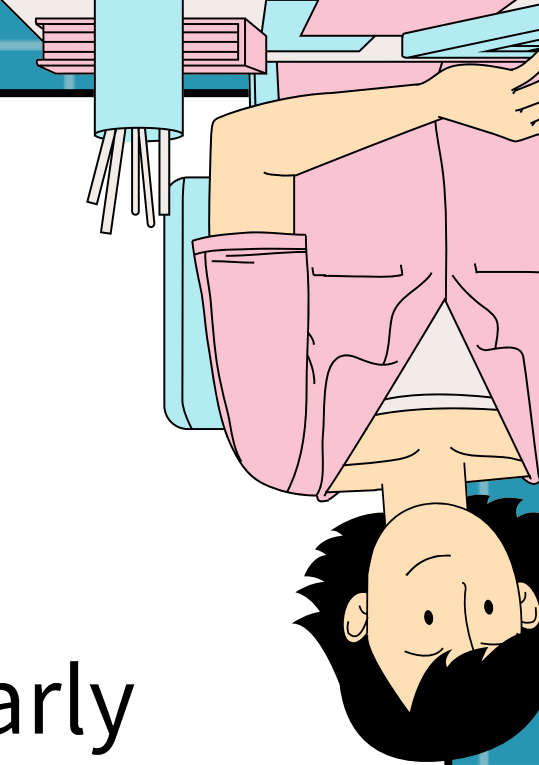
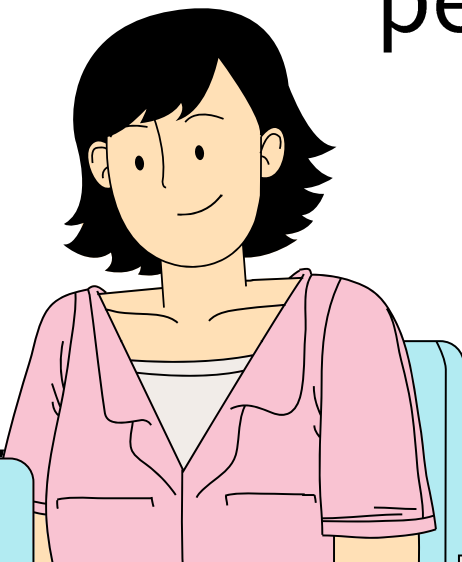
TOXICITY

	Random Over Sampling + Random Forest	Random Under Sampling + Random Forest	SMOTE + Random Forest	Random Over Sampling + Logistics Regression	Random Under Sampling + Logistics Regression	SMOTE + Logistics Regression	Random Over Sampling + XGBoost Classifier	Random Under Sampling + XGBoost Classifier	SMOTE + XGBoost Classifier	Random Over Sampling + BERT	Random Under Sampling + BERT	SMOTE + BERT
Precision	0.8727	0.8921	0.8996	0.9106	0.8916	0.9075	0.9059	0.8903	0.9067	0.9125	0.8871	0.9115
Recall	0.9129	0.7197	0.8873	0.7948	0.6590	0.7882	0.8497	0.6138	0.8524	0.8133	0.6330	0.8290
F1-score	0.8844	0.7907	0.8920	0.8418	0.7501	0.8376	0.8740	0.7170	0.8757	0.8543	0.7304	0.8638
AUC ROC	0.8793	0.8050	0.8793	0.9144	0.8236	0.8975	0.9419	0.7859	0.9381	0.9324	0.8333	0.9314

RESULT

Comparison of Data Processing Methods

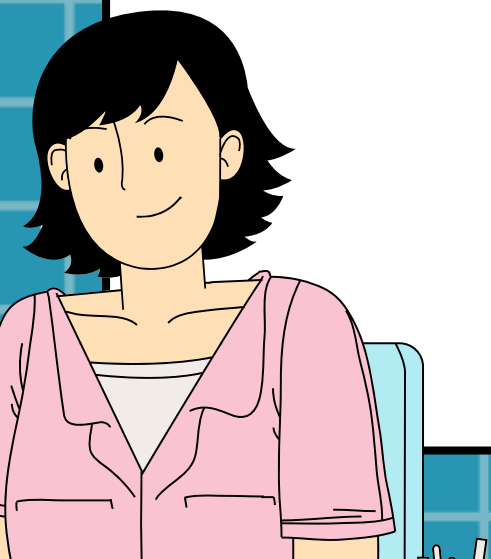
- Random Over sampling: Improved recall and F1-score, particularly enhancing detection for identity_hate, insult, and threat categories.
- Random Under sampling: Increased precision (especially for identity_hate, severe_toxic, and threat) but reduced recall and F1-score for the toxic category.
- SMOTE: Maintained a balance between precision and recall but performed slightly worse than random oversampling overall.



RESULT

Model Comparison Analysis

- XGBClassifier: Achieved the best precision in obscene, identity_hate, and threat categories.
- Random Forest: Performed exceptionally well in insult and toxic categories, with the highest overall recall.
- Logistic Regression: Demonstrated stability in severe_toxic and threat categories, particularly under random under sampling.
- BERT Model:
 - Leveraged semantic representation capabilities, maintaining stable and superior performance across all metrics (precision, recall, F1-score, ROC-AUC).
 - Achieved an AUC exceeding 0.96 in most categories, proving to be comparable to or even better than traditional models.



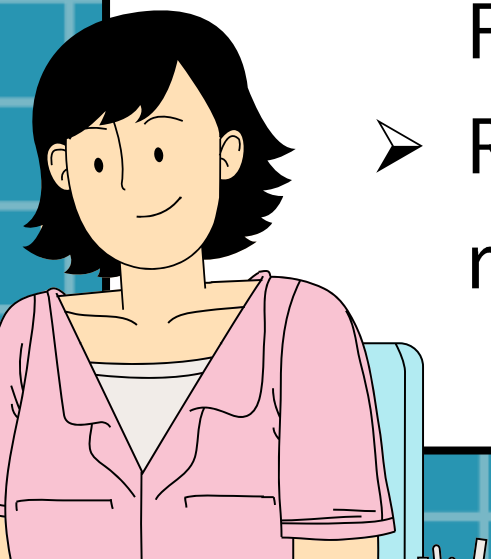
RESULT

Top Overall Performance:

- XGBoost + Random Oversampling: Achieved the highest precision and the highest AUC-ROC .
- Random Forest + Random Oversampling: Recorded the highest recall and a strong F1-score.
- Balanced Approach: BERT + SMOTE: Delivered the highest precision while maintaining high recall , F1-score , and AUC-ROC , offering the best trade-off between accuracy and coverage.

Sampling Method Comparison:

- Random Oversampling: Produced the highest overall recall and AUC-ROC, making it most effective for detecting minority classes²⁹.
- SMOTE: Struck the best balance across all metrics — precision (>0.90), recall ($0.82-0.89$), F1 (>0.87), and AUC (>0.93) — performing just below Oversampling.
- Random Undersampling: Slightly improved precision (~ 0.89) but significantly reduced recall ($0.61-0.72$), F1 ($0.71-0.79$), and AUC ($0.78-0.83$).



Reference

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THANK YOU

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